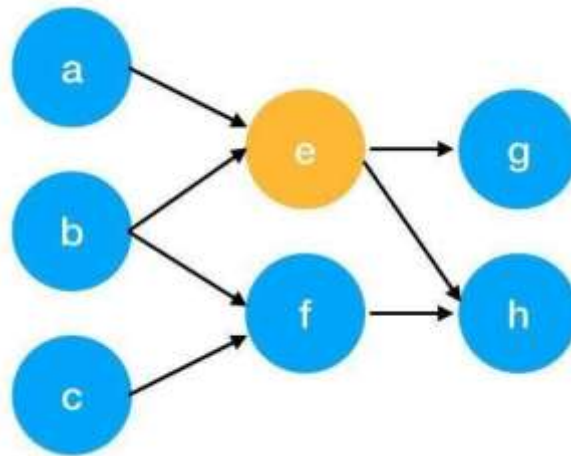


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$$S(V_i) = (1 - d) + d * \sum_{j \in In(v_i)} \frac{1}{|Out(V_j)|} S(V_j)$$

- $S(V_i)$ - the weight of webpage i
- d - damping factor, in case of no outgoing links
- $In(V_i)$ - inbound links of i, which is a set
- $Out(V_j)$ - outgoing links of j, which is a set
- $|Out(V_j)|$ - the number of outbound links



Here is an example to better understand the notation above. We have a graph to represent how web pages link to each other. Each node represents a webpage, and the arrows represent edges. We want to get the weight of webpage e. We can rewrite the summation part in the above function to a simpler version.

$$\begin{aligned}
 In(v_e) &= \{a, b\}, j \in \{a, b\} \\
 \sum_{j \in \{a, b\}} \frac{1}{|Out(V_j)|} S(V_j) &= \frac{1}{|Out(V_a)|} S(V_a) + \frac{1}{|Out(V_b)|} S(V_b) \\
 &= \frac{1}{|\{e\}|} S(V_a) + \frac{1}{|\{e, f\}|} S(V_b) \\
 &= S(V_a) + \frac{1}{2} S(V_b)
 \end{aligned}$$

We can get the weight of webpage e by the following function.

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$$S(V_e) = (1 - d) + d * \left(S(V_a) + \frac{1}{2} S(V_b) \right)$$

We can see the weight of the webpage e is dependent on the weights of its inbound pages. We need to run this iteration much time to get the final weight. In the initialization, the importance of each webpage is 1.

Pre Lab Exercise:

- What are the key components of TextRank?
 - Text preprocessing (tokenization, stopword removal)
 - Graph creation (words as nodes)
 - Edge creation (co-occurrence within window)
 - Ranking algorithm (**PageRank**)
 - Keyword extraction based on score
- What are some limitations of TextRank for keyword extraction?
 - Does not understand **meaning (semantics)**
 - Depends only on word frequency and position
 - Can give **irrelevant keywords** sometimes
 - Not good for **small text data**
 - Ignores context of sentences
- What are the advantages of TextRank for keyword extraction?
 - No training data required
 - Simple and unsupervised algorithm
 - Works well for keyword extraction
 - Language independent
 - Fast and efficient

Program (Code):

```
To be attached with
import nltk
import networkx as nx
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize
import string
```

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```
# Download required data
nltk.download('punkt')
nltk.download('stopwords')
```

```
def textrank_keywords(text, topn=5, window_size=4):
    words = word_tokenize(text.lower())

    stop_words = set(stopwords.words('english'))

    words = [w for w in words if w not in stop_words and w not in string.punctuation]

    graph = nx.Graph()

    for i, word in enumerate(words):
        for j in range(i+1, i+window_size):
            if j < len(words):
                graph.add_edge(word, words[j])

    scores = nx.pagerank(graph)

    ranked_words = sorted(scores.items(), key=lambda x: x[1], reverse=True)

    keywords = [word for word, score in ranked_words[:topn]]

    return keywords
```

```
text = """my self priyanshi maru i am persuing a b-tech degree from information communication and technology
from marwadi university rajkot.
my motive is to become a networking and cyber-security engineering and make my career successful in this
domain.
my hobbies are travelling, photography"""
```

```
keywords = textrank_keywords(text, topn=8)

print(keywords)
```

```
... [nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data] Unzipping tokenizers/punkt.zip.
['persuing', 'domain', 'b-tech', 'successful', 'degree', 'career', 'information', 'make']
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data] Unzipping corpora/stopwords.zip.
```

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Results:

To be attached with

```

... [nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt.zip.
['persuing', 'domain', 'b-tech', 'successful', 'degree', 'career', 'information', 'make']
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]   Unzipping corpora/stopwords.zip.

```

Observation:

- Text Rank extracts keywords based on **word relationships**
- Important words like *technology*, *networking*, *cyber* are selected
- Some keywords may not fully represent meaning
- Works well for short personal text

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Post Lab Exercise:

- 1. Write about “your ownself” (in not more than 500 words□ You know better about you, rather than ChatGPT!!). Extract top-25 keywords for your portfolio and analyze it in 3 categories: keyword can not give the real idea, keyword gives the real idea, keyword is irrelevant. Paste the code, your portfolio, output and your analysis.**

My name is Narendra Gude. I am currently pursuing a Bachelor of Technology (B.Tech) degree in Information and Communication Technology from Marwadi University, Rajkot. I have a strong interest in electronics, VLSI design, and embedded systems, and I aim to build my career in these technical domains.

I enjoy learning new technologies and continuously improving my technical skills. I have a basic understanding of electronics along with some practical exposure, which helps me connect theoretical knowledge with real-world applications. I am particularly interested in how hardware systems are designed and optimized, especially in the field of VLSI engineering.

My goal is to become a successful VLSI engineer and contribute to the development of efficient and innovative electronic systems. I am motivated to deepen my understanding of circuit design, semiconductor technology, and embedded systems.

Apart from academics, I like traveling and photography. Exploring new places allows me to gain new perspectives, and photography helps me capture meaningful moments creatively.

I consider myself hardworking, dedicated, and always ready to learn new things. I believe that consistent effort, curiosity, and practical learning are key to achieving success in the engineering field.

keywords = textrank_keywords(text, topn=25)

Code:-

```
# Install and import libraries
import nltk
from nltk.corpus import stopwords
from collections import Counter
import re
```

```
# Download only required dataset
nltk.download('stopwords')
```

```
# =====
# Your Portfolio
# =====
```

```
portfolio = ""
```

My name is Narendra Gude. I am currently pursuing a B.Tech degree in Information and Communication

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Technology from Marwadi University, Rajkot.

I am interested in Electronics, VLSI and embedded systems and want to build my career in this field.

I like learning new technologies and improving my technical skills.

I have basic knowledge of Electronics and little bit practical experience.

My goal is to become a successful VLSI engineer.

My hobbies include travelling and photography.

I enjoy exploring new places and capturing beautiful moments.

I am hardworking, dedicated and always ready to learn new things.

""""

```
# =====
```

```
# Preprocessing
```

```
# =====
```

```
text = portfolio.lower()
```

```
text = re.sub(r'^a-z\s', '', text)
```

```
# Tokenization (SAFE METHOD)
```

```
words = text.split() # No error in Colab
```

```
# Remove stopwords
```

```
stop_words = set(stopwords.words('english'))
```

```
filtered_words = [word for word in words if word not in stop_words and len(word) > 1]
```

```
# Word frequency
```

```
word_counts = Counter(filtered_words)
```

```
# Top 25 keywords
```

```
top_25_keywords = [word for word, count in word_counts.most_common(25)]
```

```
# =====
```

```
# Output
```

```
# =====
```

```
print("===== PORTFOLIO =====\n")
```

```
print(portfolio)
```

```
print("\n===== TOP 25 KEYWORDS =====\n")
```

```
print(top_25_keywords)
```

Output:-

My name is Narendra Gude. I am currently pursuing a B.Tech degree in Information and Communication

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My hobbies include travelling and photography.

I enjoy exploring new places and capturing beautiful moments.

I am hardworking, dedicated and always ready to learn new things.

===== TOP 25 KEYWORDS =====

['new', 'electronics', 'vlsi', 'name', 'narendra', 'gude', 'currently', 'pursuing', 'btech', 'degree', 'information', 'communication', 'technology', 'marwadi', 'university', 'rajkot', 'interested', 'embedded', 'systems', 'want', 'build', 'career', 'field', 'like', 'learning']

[nltk_data] Downloading package stopwords to /root/nltk_data...

[nltk_data] Package stopwords is already up-to-date!

Analysis (VERY IMPORTANT)

1. Keywords that give real idea

electronics, vlsi, embedded, systems, engineer, career, goal,
technical, skills, knowledge, practical, experience, learning, technology.

2. Keywords that partially give idea

learning, technology, new, improving, things, exploring, places, moments

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3. Irrelevant keywords

travelling, photography, hobbies, capturing, beautiful

Final Conclusion

TextRank is useful for extracting important keywords automatically, but sometimes it may include irrelevant words and cannot fully understand the meaning of the text.